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Task: Machine Learning Research Assignment

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PART A: What is Machine Learning?

A1: What is machine learning?

Machine learning is a field of artificial intelligence in which computer systems learn patterns from data and use those patterns to make predictions or decisions. In traditional programming, a programmer writes explicit rules for the computer to follow. In machine learning, an algorithm learns a pattern from examples.

For example, a traditional system might flag a transaction whenever its value exceeds a fixed amount. A machine learning system could learn from historical transactions labelled as fraudulent or legitimate and use several features to estimate whether a new transaction is fraudulent.

A2: How does machine learning relate with Artificial intelligence, deep learning, and data science?

Artificial intelligence (AI) is the broad field concerned with computer systems performing tasks associated with human intelligence. Machine learning is one approach within AI that allows systems to learn patterns from data. Deep learning is a branch of machine learning that uses multi-layer neural networks to learn complex patterns.

Data science is a broader, overlapping discipline concerned with extracting useful knowledge from data through collection, cleaning, analysis, visualisation and, where appropriate, predictive modelling.

A3: Discuss the core terms:

Dataset- The complete table containing records of houses and their selling prices.

Feature - An input variable, such as floor area, suburb or number of bedrooms.

Label or target- The outcome we want to predict, such as selling price.

Model- The learned mathematical relationship between the house features and selling prices.

Training- The process of using historical examples to learn that relationship.

Prediction- Using the trained model to estimate the selling price of a new house

A4: Find 3 real world uses of machine learning in South Africa or elsewhere in Africa

1. Fraud detection: A financial institution can use transaction amounts, timing, location and transaction history to estimate the likelihood of fraud. The institution benefits from earlier detection and reduced losses.
2. Agricultural disease detection: Images of crops, weather information and farm observations can help a model identify potential crop diseases. Farmers can benefit from earlier investigation and treatment.
3. Telecommunications fault prediction: Network performance measurements, error logs and historical equipment failures can help predict network faults. Telecommunications providers can prioritise maintenance and reduce service interruptions.

These are illustrative applications, not claims that every named application is currently deployed by a particular African organisation. For your final report, strengthen this section by researching documented African examples.

A5: The machine learning workflow

The typical workflow begins with defining the problem and the outcome to predict. Next, relevant data is collected and explored. The data is cleaned, missing values are handled, categories are encoded and numeric features are scaled when necessary. An appropriate algorithm is selected and trained. Its performance is then evaluated on data that was held back from training. If the results meet the required standard, the model can be deployed and monitored.

Data preparation often takes substantial time because real-world datasets may contain missing values, inconsistent formats, duplicates, unusual observations and irrelevant columns. Poor-quality inputs can lead to misleading predictions.

Part B: Supervised versus unsupervised learning

B1. Supervised learning

Supervised learning uses labelled training data, meaning the correct outcome is known for the training examples. The model learns a relationship between input features and the target. An example is training a model on SMS messages already labelled as spam or not spam.

B2. Unsupervised learning

Unsupervised learning works with data without predefined target labels. The algorithm looks for patterns or structure in the observations. For example, a retailer can group customers by spending patterns without knowing the customer segments in advance.

B3: Supervised learning vs Unsupervised learning:

Aspect	Supervised learning	Unsupervised learning
Goal	Predict a known target	Discover patterns or structure
Data required	Labelled data	Data without target labels
Typical tasks	Classification and regression	Clustering and dimensionality reduction
Algorithms	Logistic Regression, Decision Trees, Linear Regression	K-Means, PCA

Evaluation	Metrics such as accuracy, F1-score, MAE or RMSE, depending on the task	Measures such as silhouette score, explained variance or domain-specific usefulness
Real-world example	Predict whether a transaction is fraudulent using historical labels	Group customers according to purchasing behaviour

B4. Classification versus regression

Classification predicts a category, such as whether an SMS is spam or not spam. Regression predicts a numerical value, such as a learner's final examination mark or a property's selling price. Both are supervised learning tasks when trained using known target values.

B5. Clustering versus dimensionality reduction

Clustering groups observations that are similar according to selected features. For example, a retailer could group products with similar sales patterns.

Dimensionality reduction represents data using fewer dimensions while retaining important information. For example, PCA can reduce many correlated numeric variables to a smaller set of components, making data easier to analyse or visualise.

B6. Classify the five scenarios

Scenario	Answer	Reason

(a) Predicting a house's selling price	Supervised regression	Historical selling prices provide numerical targets.
(b) Grouping customers by spending habits	Unsupervised clustering	The groups are not defined beforehand.
(c) Deciding whether an SMS is spam	Supervised classification	The target is a category, such as spam or not spam
(d) Predicting a learner's final mark	Supervised regression	The target is a numerical mark.
(e) Finding groups of products with similar sales patterns	Unsupervised clustering	The objective is to discover similar groups.

B7. The six scikit-learn classes

Class	Learning type	Purpose	Module path
LinearRegression	Supervised	Predicts a continuous numerical value using a linear relationship.	sklearn.linear_model.LinearRegression

LogisticRegression.	Supervised	Predicts class membership using a classification model.	sklearn.linear_model.LogisticRegression
DecisionTreeClassifier	Supervised	Predicts classes using decision rules arranged as a tree.	sklearn.tree.DecisionTreeClassifier
KNeighborsClassifier	Supervised	Classifies an observation using nearby labelled examples.	sklearn.neighbors.KNeighborsClassifier
KMeans	Unsupervised	Groups observations into a specified number of clusters	sklearn.cluster.KMeans
PCA	Unsupervised	Reduces dimensionality by creating principal components.	sklearn.decomposition.PCA

Part C: Written research answers

The notebook demonstrates the practical skills.

C1. Exploring and cleaning data

When opening a new dataset, I would inspect its dimensions, column names, data types, missing values, duplicate rows and descriptive statistics. In pandas, df.info() summarises the structure and non-null counts, df.describe() provides descriptive statistics, df.isnull().sum() counts missing values, and df.duplicated() identifies repeated rows.

If a numeric column contains numbers stored as text, I would investigate the values before converting the column using astype() or pd.to_numeric(). Invalid values should be identified rather than silently discarded.

C2. Handling missing values

Missing values can be handled by removing rows or columns, or by replacing missing entries through imputation. Dropping rows may be reasonable when only a few observations are affected and the missingness is not problematic. However, on a small dataset it may remove valuable information or introduce bias.

Mean imputation can be suitable for reasonably symmetric numeric data without influential outliers. Median imputation is often preferable for skewed data because it is less affected by extreme values. Mode imputation can be used for categorical features, while a constant may be appropriate when it has a meaningful interpretation.

The SimpleImputer class in sklearn.impute provides a consistent way to perform imputation. In a machine learning workflow, its replacement values should be learned from training data only.

C3. Encoding categorical data

Machine learning algorithms often require numerical input. Encoding converts categories into numerical representations.

Nominal categories have no natural order, such as provinces or colours. Ordinal categories have an order, such as low, medium and high. OneHotEncoder creates binary columns for nominal categories, avoiding an artificial ranking. OrdinalEncoder maps categories to numerical values in an explicitly defined order.

LabelEncoder is intended primarily for target labels rather than ordinary input features. One-hot encoding can create a large number of columns when a feature has hundreds of unique categories. This can increase memory usage and computational cost. Depending on the dataset, grouping rare categories or using another suitable encoding technique may help.

C4. Scaling numeric features

Features measured on different scales can affect algorithms that use distances or regularisation. For example, salary values may be much larger than age values, causing salary to dominate distance calculations if the features are not appropriately scaled.

Standardisation uses the mean and standard deviation to centre and rescale values. Min-max normalisation maps the observed feature range to a specified range, commonly 0 to 1. RobustScaler uses statistics such as the median and interquartile range, making it less sensitive to extreme values.

KNN, SVM and many logistic regression workflows commonly benefit from scaling. Decision trees and random forests generally do not require it because their splits depend on feature thresholds rather than distances between scaled observations.

C5. Train-test splitting and data leakage

A train-test split evaluates whether a model generalises to observations it did not use during training. An 80/20 split is a common starting point, although the appropriate split depends on the amount and structure of data.

The random_state parameter makes a random split reproducible. stratify helps preserve class proportions for classification tasks.

Data leakage occurs when information that should be unavailable during training influences model fitting or evaluation. For example, fitting a scaler using the entire dataset allows the test set to influence the training process.

fit() learns the necessary values, transform() applies those learned values, and fit_transform() performs both operations. Fit preprocessing on training data and then transform the training and test sets using the learned preprocessing.

C6. Detecting outliers

Outliers are observations that differ substantially from the rest of the data. They can be identified using boxplots, the interquartile range (IQR) rule or z-scores.

Under the common IQR rule, observations below \(Q_1-1.5(IQR)\) or above \(Q_3+1.5(IQR)\) are flagged as potential outliers. These are screening thresholds, not proof that an observation is incorrect.

An age of 250 years might indicate a data-entry error. A very large financial transaction, however, may be genuine and important for fraud detection. Outliers should therefore be investigated before deciding whether to correct, retain, cap or remove them.

C7. Feature engineering

Feature engineering involves creating useful input variables from existing information. For example, a date column can be used to derive the month or day of the week, while a birth year can be used to estimate age if the relevant dates are available.

A correlation matrix helps identify linear relationships between numeric features. Highly correlated features may contain overlapping information, although correlation alone does not prove that a feature is useless or that it causes another variable to change. The decision to retain or remove features should also consider the prediction objective, domain knowledge and model performance.

Part D: Further Research

D1. Overfitting and underfitting

Overfitting and underfitting are two common problems in machine learning that affect how well a model makes predictions on new data. Overfitting happens when a model learns the training data too closely, including random patterns or noise that may not occur in other datasets. As a result, it may perform very well on the training data but poorly on unseen data. For example, a model predicting house prices might memorise the specific houses in its training dataset instead of learning general patterns that influence house prices.

Underfitting occurs when a model is too simple to learn the important patterns in the data. It usually performs poorly on both training and test data. For example, using a simple straight-line model to describe a complicated relationship between house size, location and price may not capture the relationship accurately.

Scikit-learn explains that model complexity is related to the balance between bias and variance. A model with high bias may underfit, while a model that is too sensitive to its training data may overfit.

scikit-learn 1.9.1 documentation

To reduce overfitting, a data scientist can use cross-validation, regularization, simpler models or more representative training data. Underfitting may be reduced by selecting a more suitable model, adding useful features or allowing the model to learn more complex relationships. The goal is to find a model that learns meaningful patterns and generalises well to new data.

D2. Measuring a model's performance

Model evaluation is the process of measuring how well a machine learning model performs on data. It helps data scientists decide whether a model is suitable for its intended purpose. The correct evaluation metric depends on the type of problem

being solved. Scikit-learn provides different metrics for classification, regression and clustering tasks.

GitHub

For classification, a model predicts categories, such as whether a customer will leave a company. Accuracy measures the proportion of predictions that are correct. Precision measures how many predicted positive cases are actually positive, while recall measures how many actual positive cases the model identifies. The F1-score combines precision and recall into one measure. Accuracy alone can be misleading when one class is much more common than another, such as when detecting rare cases of fraud.

For regression, a model predicts numerical values, such as house prices. Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values. Mean Squared Error (MSE) squares the errors before averaging them, giving larger errors greater influence. The R-squared score indicates how well the model explains variation in the target variable, although it should not be used alone to judge model quality.

A model should be evaluated using data that was not used to train it. The results should also be interpreted in relation to the real problem. For example, a fraud detection model may need high recall to identify more fraudulent transactions, even if that creates additional false alarms.

D3. Cross-validation

Cross-validation is a technique used to estimate how well a machine learning model may perform on unseen data. It is useful because a model's performance can depend on which observations are included in its training and testing sets. Scikit-learn describes several cross-validation methods, including K-fold cross-validation.

scikit-learn 1.9.1 documentation

In K-fold cross-validation, the dataset is divided into K smaller groups called folds. For example, in five-fold cross-validation, the data is divided into five parts. The model is trained five times. Each time, four folds are used for training and the

remaining fold is used for validation. Every fold is used once for validation. The five evaluation scores can then be averaged to provide an overall estimate of model performance.

One advantage of cross-validation is that it uses different portions of the dataset for training and validation, giving a more informative assessment than relying on a single split. It can also help compare models and select suitable hyperparameters. However, it requires additional computing time because the model is trained multiple times.

Cross-validation must be applied carefully. For example, when scaling numerical features or filling missing values, the preprocessing steps should be fitted only on the training portion of each fold. Otherwise, information from the validation data could influence the model, causing data leakage. A scikit-learn pipeline can help prevent this problem.

GitHub

Cross-validation is therefore a valuable tool for model selection, but a separate final test set can still be useful for an independent evaluation after the model has been selected.

D6. K-means clustering

K-means is an unsupervised machine learning algorithm that groups similar data points into clusters. Unlike supervised learning, it does not require a known target label for every observation. Instead, it identifies groups based on similarities in the input features. According to the scikit-learn documentation, K-means aims to form clusters by minimizing the sum of squared distances between data points and their assigned cluster centers, known as centroids.

scikit-learn 1.9.1 documentation

The algorithm works by selecting the number of clusters, represented by K, and initializing K centroids. Each data point is assigned to the nearest centroid. The algorithm then calculates updated centroids based on the mean of the points

assigned to each cluster. These steps repeat until the centroids stop changing significantly, or a stopping condition is reached.

For example, a retailer could use K-means to group customers according to their purchasing frequency and total spending. The resulting clusters might reveal groups of frequent high-spending customers, occasional shoppers, and customers who spend less. The retailer could then use these patterns to inform marketing strategies.

One advantage of K-means is that it is relatively simple to understand and can work efficiently on large datasets. However, the number of clusters must be selected in advance. Results can also be affected by outliers, feature scales, and the initial centroid positions. Since K-means relies on distances, numerical features may need scaling before clustering. It also works best when the data has cluster shapes that fit its assumptions, rather than irregularly shaped groups.

K-means can support customer segmentation and exploratory data analysis, but its clusters must be interpreted in the context of the business problem.

D9. Machine learning pipelines

A machine learning pipeline is a sequence of steps that processes data and trains a model in an organised workflow. A pipeline can combine preprocessing operations, such as filling missing values and scaling numerical features, with a machine learning algorithm. Scikit-learn provides a Pipeline tool that allows these steps to be fitted and used together.

scikit-learn 1.9.1 documentation

For example, a pipeline for predicting whether a customer will leave a company could first replace missing numerical values, then scale numerical features, and finally train a classification model. Instead of managing each operation separately, the data scientist can fit the complete pipeline using the training data and use it to make predictions on new data.

Pipelines have several advantages. First, they make code easier to organise, reuse and maintain. Second, they help ensure that the same preprocessing steps are

applied consistently during training and prediction. Third, they can reduce data leakage, which occurs when information from validation or test data influences the model during training. Scikit-learn explains that pipelines help avoid this by fitting preprocessing steps within the appropriate training process.

scikit-learn 1.9.1 documentation

Pipelines are especially useful when using cross-validation or tuning model parameters because preprocessing and modelling can be handled as one combined workflow. However, the pipeline still needs to be designed correctly: the target variable must be separated from the input features, the data must be split appropriately, and each transformation must suit the data.

Overall, pipelines support a more consistent and reliable machine learning process. They are particularly valuable when projects involve several preprocessing steps and need to be repeated or deployed.

Reference list (APA 7th edition)

Use the following references at the end of your report. These are official scikit-learn documentation pages, which are relevant technical sources for the five topics.

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